



2026 BASIC ABSTRACT ALGEBRA STUDY

# PRE-COURSE SELF ASSESSMENT

## SOLUTION - Q6

**Version**

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**Read this carefully before moving on.**

In the following questions, *prove or disprove* means you need to either show a correct logical proof of a given statement or provide at least one counterexample that disproves the statement. Describe each deduction step by step, citing the relevant definition or fact at each step.

As a tiny example of a proof answer, consider the following statement.

For any integers  $a$  and  $b$ , if  $a$  and  $b$  are both even, then so is  $a + b$ .

The above statement is true.

*Proof.* Assume  $a$  and  $b$  are even integers. By the definition of an even integer, there exist integers  $k$  and  $l$  such that  $a = 2k$  and  $b = 2l$ . Then  $a + b = 2k + 2l = 2(k + l)$ , and since  $k + l$  is an integer,  $a + b$  is even by the same definition.  $\square$

As a tiny example of a disproof answer, consider the following statement.

For any integers  $a$  and  $b$ , if  $a + b$  is even, then  $a$  and  $b$  are both even.

The above statement is false.

*Proof.* Take  $a = 1$  and  $b = 1$ . Both are odd, but  $a + b = 2$  is even. Hence the statement is false.  $\square$

It is highly encouraged to write down full sentences like the above samples, rather than just throwing out a few symbols or numbers without explaining why.

**Note 1**

The following symbols will be used throughout the following questions.

- $\mathbb{Z}$ : A set of all integers.
- $\mathbb{Z}^+$ : A set of all positive integers. i.e.,  $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$ .
- $\mathbb{Z}_n$ : A set of non-negative integers from 0 to  $n - 1$  for some positive integer  $n$ . For instance,  $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ .
- $\mathbb{Q}$ : A set of all rational numbers.
- $\mathbb{R}$ : A set of all real numbers.
- $\mathbb{R}^+$ : A set of all positive real numbers.
- $\mathbb{C}$ : A set of all complex numbers.

**Definition 2 (Extensionality)**

Two sets  $X$  and  $Y$  are said to be equal if and only if  $X$  is a subset of  $Y$  and  $Y$  is a subset of  $X$ . This is called the *axiom of extensionality*. If two sets  $X$  and  $Y$  are equal, we write  $X = Y$ .

**Q1.**

**Note 3.**  $A \setminus B$  denotes the set of all elements in  $A$  that are not in  $B$ .

Which of the following is distinct from the others?

- ☐ 1.  $\{-x \mid x \in \mathbb{R}\}$ 
☐ 2.  $\{xx \mid x \in \mathbb{R}\}$ 
☐ 3.  $\{xy \mid x, y \in \mathbb{R}\}$
- ☐ 4.  $\{xxx \mid x \in \mathbb{R}\}$ 
☐ 5.  $\{\frac{x}{y} \mid x \in \mathbb{R}, y \in \mathbb{R} \setminus \{0\}\}$

## Q2.

**Note 4.** The sum of two integers is always an integer.

Show that  $\{x + y \mid x, y \in \mathbb{Z}\} = \mathbb{Z}$  using extensionality. (Definition 2)

**Hint 5.** First, show that  $\{x + y \mid x, y \in \mathbb{Z}\} \subset \mathbb{Z}$ . Then, show that  $\mathbb{Z} \subset \{x + y \mid x, y \in \mathbb{Z}\}$ . In the latter case, it is enough to show that for any integer, there exist at least a pair of integers such that their sum gets equal to that integer.

### Definition 6 (Size of finite sets)

A size of a finite set  $X$  is the number of elements it contains, and denoted as  $|X|$ .

## Q3.

Let  $A$  be a set of all prime numbers that are even. What is the value of  $|A|$ ?

- ☐ 1. 0
 ☐ 2. 1
 ☐ 3. 2
 ☐ 4. 3
 ☐ 5. 4

## Q4.

Let  $A = \{n + m \mid n, m \in \mathbb{Z}_{20261212}\}$ . What is the value of  $|A|$ ?

**Hint 7.**  $|\{n + m \mid n, m \in \mathbb{Z}_2\}| = |\{0 + 0, 0 + 1, 1 + 0, 1 + 1\}| = |\{0, 1, 2\}| = 3$ .

Answer: \_\_\_\_\_

## Q5.

Let  $T$  be the set of all real numbers except 0 and 1. Define two functions  $f, g : T \rightarrow T$  as  $f(x) = x^{-1}$  and  $g(x) = 1 - x$ , respectively.

Let  $S$  be the set of all possible functions that can be created by composing  $f$  and  $g$  any number of times in any order, such as  $f, g, g \circ f, f \circ g, g \circ g \circ f \circ f, f \circ g \circ f, f^{20261212}$ , etc.

What is the value of  $|S|$ ?

Answer: \_\_\_\_\_

Q6.

Let  $\text{Di}_6 = \{\square, \square, \square, \square, \square, \square\}$  be a set of all possible faces of a six-sided die.

How many distinct ordered pairs can be made from  $\text{Di}_6$ ?

- ☒ 1. 36     
 ☐ 2. 21     
 ☐ 3. 6     
 ☐ 4. 30     
 ☐ 5. 12

Keep in mind that the order matters in an ordered pair. The total number of distinct pairs out of 6 elements is  $6^2 = 36$ .

Therefore, the first choice is the correct answer.

### Definition 8 (Product of sets)

Let  $X, Y$  be two sets. A *product*  $X \times Y$  between them is a set of all ordered pairs  $(x, y)$  where  $x \in X$  and  $y \in Y$ .

Q7.

Let  $X = \{10, 20, 30\}$  and  $Y = \{\leftarrow, \rightarrow\}$ . Find the same set as  $Y \times X$ .

- ☐ 1.  $\{(10, 10), (20, 20), (30, 30), (\leftarrow, \leftarrow), (\rightarrow, \rightarrow)\}$   
☐ 2.  $\{(10, \leftarrow), (10, \rightarrow), (20, \leftarrow), (20, \rightarrow), (30, \leftarrow), (30, \rightarrow)\}$   
☐ 3.  $\{(\leftarrow, 20), (20, \leftarrow), (\leftarrow, 20), (20, \leftarrow), (30, \rightarrow), (\rightarrow, 30)\}$   
☐ 4.  $\{(10, \rightarrow), (20, \rightarrow), (30, \rightarrow), (10, \leftarrow), (20, \leftarrow), (30, \leftarrow)\}$   
☐ 5.  $\{(\leftarrow, 20), (\leftarrow, 30), (\leftarrow, 10), (\leftarrow, 20), (\rightarrow, 10), (\rightarrow, 20), (\rightarrow, 30)\}$

### Definition 9 (Power set)

A set of all subsets of a set  $X$  is called a *power set* of  $X$ , and denoted as  $\mathcal{P}(X)$ .

Q8.

Which of the following is the same as  $\mathcal{P}(\{a, 1, *\})$ , if  $a, 1$ , and  $*$  are all distinct?

- ☐ 1.  $\{\{\emptyset, \{a\}, \{1\}, \{*\}, \{a, 1\}, \{1, *\}, \{a, *\}, \{a, 1, *\}\}\}$   
☐ 2.  $\{\{*, 1, a\}, \{*, a\}, \{*, 1\}, \{1, a\}, \{*\}, \{a\}, \{1\}\}$   
☐ 3.  $\{\emptyset, \{1\}, \{a\}, \{*\}, \{(a, 1)\}, \{(1, *)\}, \{(a, *)\}, \{(1, a, *)\}\}$   
☐ 4.  $\{\{*, 1, a\}, \{a\}, \{a, 1, a\}, \{*, *\}, \{1\}, \emptyset, \{1, *\}, \{a\}, \{a, *, *\}\}$   
☐ 5.  $\{\emptyset, \{a\}, \{1\}, \{*\}, \{a, 1\}, \{1, a\}, \{1, *\}, \{*, 1\}, \{a, 1, *\}\}$

**Q9.**What is the value of  $|\mathcal{P}(\mathbb{Z}_2 \times \mathbb{Z}_5)|$ ?

Answer: \_\_\_\_\_

**Q10.**Prove or disprove that the following holds for any two sets  $A$  and  $B$ .

$$(A \setminus B) \cap (B \setminus A) = \emptyset$$

**Q11.**Prove or disprove that the following holds for any set  $X$ .

$$\mathcal{P}(X) \cap X = \emptyset$$

**Definition 10 (Singleton)**A set is called a *singleton* if it has only one element.**Q12.**

Prove or disprove the following statement.

A set is a singleton if and only if it has only one proper subset.

**Definition 11 (Injection, Surjection, and Bijection)**

A function  $f : X \rightarrow Y$  is said to be *injective* or called an *injection* if for every  $x, y \in X$ , if  $x \neq y$  then  $f(x) \neq f(y)$ . A function is said to be *surjective* or called a *surjection* if for every  $y \in Y$ , there exists an  $x \in X$  such that  $f(x) = y$ . A function is said to be *bijective* or called a *bijection* if it is both an injection and a surjection.

**Q13.**

For a set  $X$ , think of another set  $X/=$  that is a set of all singletons from all elements of  $X$ . For instance, if  $X$  is  $\{a, b, c\}$ , then  $X/=$  becomes  $\{\{a\}, \{b\}, \{c\}\}$ .

Prove or disprove that there is a bijection between  $X$  and  $X/=$ .

**Definition 12 (Equality of functions)**

Two functions with the same domain and same codomain are said to be *equal* if and only if they map every element of the domain to the same element of the codomain.

In other words, if  $f, g : X \rightarrow Y$  are functions with the same domain and codomain, then  $f = g$  if and only if  $f(x) = g(x)$  for every  $x \in X$ .

**Q14.**

Show that for any singleton, there is always a unique function from any set to it.

**Hint 13.** Let  $\{*\}$  be a singleton and  $X$  be any set, and consider a function  $f : X \rightarrow \{*\}$ . First show that such function always exists, and then prove that if there is a function  $f' : X \rightarrow \{*\}$ , then  $f = f'$ .

**Q15.**

**Note 14.** An induction here refers to the mathematical induction.

Let  $A_i$  be a singleton for  $i = 1, \dots, n$  for a positive integer  $n$ . Show by induction that  $A_1 \times \dots \times A_n$  is also a singleton.

**Q16.**

Let  $X, Y$  be finite sets. How many functions  $f : X \rightarrow Y$  are there from  $X$  to  $Y$ ?

- ☐ 1.  $|Y|^{|X|}$ 
☐ 2.  $2^{|X| \cdot |Y|}$ 
☐ 3.  ${}_{|Y|}P_{|X|}$ 
☐ 4.  $|X|^{|Y|}$ 
☐ 5.  $|X|! \cdot |Y|!$

**Q17.**

Prove or disprove that the following statement holds for any finite sets  $X, Y$  and  $Z$ .

If the number of bijections from  $X$  to  $Y$  is the same as the number of bijections from  $X$  to  $Z$ , then  $|Y| = |Z|$ .

**Q18.**

Prove or disprove that if  $f$  and  $g$  are injective, then  $g \circ f$  is injective for any functions  $f$  and  $g$ .

**Q19.**

Prove or disprove that if  $f \circ g = f \circ h$ , then  $g = h$  for any functions  $f : Y \rightarrow Z, g, h : X \rightarrow Y$ .

**Q20.**

Prove or disprove the following statement.

For any finite set  $X$  and  $Y$  where  $|X| = |Y|$ , a function  $f : X \rightarrow Y$  is injective if and only if it is surjective.

**Hint 15.** Use the pigeonhole principle.

**Q21.**

**Note 16.**  $\ln(x)$  denotes the natural logarithm of  $x$ .

Prove or disprove that  $\ln : \mathbb{R}^+ \rightarrow \mathbb{R}$  is bijective.

**Q22.**

Which of the following is the same as  $(-i)^{4443}$ ?

- ☐ **1.**  $(-i)^{666}$ 
☐ **2.**  $i^{321}$ 
☐ **3.**  $i^{123}$ 
☐ **4.**  $(-i)^{4812}$ 
☐ **5.**  $(-i)^{333}$

**Q23.**

How many trailing zeros are there in the decimal representation of  $629!$ ?

**Hint 17.** There are two trailing zeros in  $11!$ , as  $11! = 39916800$ .

Answer: \_\_\_\_\_

**Q24.**

**Note 18.** For integers  $a$  and  $b$ ,  $a \perp b$  denotes that there are no common divisors of  $a$  and  $b$  other than 1. It is equivalent to saying that  $\gcd(a, b) = 1$ . In contrast,  $a \not\perp b$  denotes that there are common divisors other than 1 between  $a$  and  $b$ .

**(a)** Prove or disprove that the following statement holds for any integers  $a$ ,  $b$  and  $c$ .

If  $a \perp b$  and  $b \perp c$ , then  $a \perp c$ .

**(b)** Prove or disprove that the following statement holds for any integers  $a$ ,  $b$  and  $c$ .

If  $a \nmid b$  and  $b \nmid c$ , then  $a \nmid c$ .

Q25.

(a) Prove or disprove that the following statement holds for any integers  $a$ ,  $b$  and  $c$ .

If  $a$  divides  $b$  and  $b$  divides  $c$ , then  $a$  divides  $c$ .

(b) Prove or disprove that the following statement holds for any integers  $a$ ,  $b$  and  $c$ .

If  $a$  does not divide  $b$  and  $b$  does not divide  $c$ , then  $a$  does not divide  $c$ .

### Lemma 19 (Euclid's lemma)

For integers  $a$ ,  $b$ ,  $c$ , if  $a \mid b$  and  $a$  divides  $bc$ , then  $a$  divides  $c$ .

Q26.

**Note 20.** For a rational number  $r$ , an irreducible form of  $r$  is a representation  $r = \frac{p}{q}$  such that  $p \in \mathbb{Z}$ ,  $q \in \mathbb{Z}^+$ , and  $p \perp q$ .

Define  $f : \mathbb{Q} \rightarrow \mathbb{Z} \times \mathbb{Z}^+$  to be the rule that sends each  $r \in \mathbb{Q}$  to  $(p, q)$ , where  $\frac{p}{q}$  is an irreducible form of  $r$ .

(a) Show that  $f$  is a well-defined function.

**Hint 21.** First, show that every rational number admits an irreducible form. Then, show that if  $\frac{p}{q}$  and  $\frac{p'}{q'}$  are both irreducible forms of the same rational number, then  $p = p'$  and  $q = q'$ .

(b) Prove or disprove that  $f$  is injective.

(c) Prove or disprove that  $f$  is surjective.

Q27.

Show by induction that an equation of the following form has a unique complex solution for each  $n \in \mathbb{Z}^+$ .

$$x^n = 0$$

**Hint 22.** If  $ab = 0$ , then either  $a$  or  $b$  is zero.

Q28.

Let  $\mathbf{X}$  be a  $m \times m$  matrix with complex entries and  $\mathbf{0}$  be the zero matrix. Prove or disprove that an equation of the following form has a unique solution for each  $n, m \in \mathbb{Z}^+$ .

$$\mathbf{X}^n = \mathbf{0}$$

**Q29.**

Let  $\mathbf{A}$  be a  $n \times n$  matrix with complex entries. Prove or disprove that the following equation has a unique solution for each  $n \in \mathbb{Z}^+$ .

$$\mathbf{A} = -\mathbf{A}$$

**Q30.**

Prove or disprove that for any positive integer  $n \in \mathbb{Z}^+$ , an equation of the following form has at most two distinct solutions in  $\mathbb{Z}_n$ .

$$x^2 \equiv 1 \pmod{n}$$

**Hint 23.** By ‘at most two distinct’ solutions, we mean that the number of solutions can be zero, one, or two, but not more than two.

**Q31.**

Given any  $a \in \mathbb{Z}_n$  such that  $a \perp n$ , prove or disprove that an equation of the following form has a unique solution in  $\mathbb{Z}_n$  for each  $n \in \mathbb{Z}^+$ .

$$ax \equiv 1 \pmod{n}$$

**Hint 24.** See **Q20**. Can you find a relevant bijection on  $\mathbb{Z}_n$ ?

**Q32.**

**(a)** How many solutions does the equation below have in  $\mathbb{Z}_{12121212}$ ?

$$1212x \equiv 4848 \pmod{12121212}$$

**Hint 25.** Try considering similar questions on  $\mathbb{Z}_3$ ,  $\mathbb{Z}_8$ , or any small  $\mathbb{Z}_n$  and solving them. Can you find a pattern?

Answer: \_\_\_\_\_

**(b)** How many solutions does the equation below have in  $\mathbb{Z}_{20261212}$ ?

$$1212x \equiv 2026 \pmod{20261212}$$

Answer: \_\_\_\_\_

(c) How many solutions does the equation below have in  $\mathbb{Z}_{20262121}$ ?

$$1212x \equiv 2026 \pmod{20262121}$$

**Hint 26.** See [Q31](#).

Answer: \_\_\_\_\_

### Q33.

Let  $\mathbb{Z}^{\mathbb{Z}}$  be a set of all functions from  $\mathbb{Z}$  to  $\mathbb{Z}$  and define a function  $f \in \mathbb{Z}^{\mathbb{Z}}$  by  $f(x) = 2x$ . We denote the identity function on  $\mathbb{Z}$  by  $1_{\mathbb{Z}}$ .

Prove or disprove that the following equation has a unique solution in  $\mathbb{Z}^{\mathbb{Z}}$ . In other words, there is a unique function  $x : \mathbb{Z} \rightarrow \mathbb{Z}$  such that the following equation holds.

$$x \circ f = 1_{\mathbb{Z}}$$

### Q34.

Let  $\Gamma$  be an arbitrary finite set of invertible  $n \times n$  matrices with complex entries with  $n \in \mathbb{Z}^+$ . Suppose that  $\Gamma$  is closed under matrix multiplication. In other words, for any  $A, B \in \Gamma$ , the product  $AB$  is also in  $\Gamma$ .

Let  $\Psi$  be the sum of all matrices in  $\Gamma$ .

$$\Psi = \sum_{\Omega \in \Gamma} \Omega$$

Show that for any matrix  $\Lambda \in \Gamma$ , the equation  $\Lambda\Psi = \Psi$  holds.

**Hint 27.** See [Q20](#).

End.

You may now submit your work.

## Evaluation sheet

For each question so far, evaluate it based on the following criteria:

- If you encountered new concepts or new words for the first time, mark ✓ in the New concepts column.
- If you failed to understand the question, mark × in the Evaluation column.
- If you understood the question, but didn't know how to solve it, mark ? in the Evaluation column.
- If you understood the question and are sure you have solved it, mark ○ in the Evaluation column.

| No. | New concepts | Evaluation | No. | New concepts | Evaluation |
|-----|--------------|------------|-----|--------------|------------|
| Q1  |              |            | Q18 |              |            |
| Q2  |              |            | Q19 |              |            |
| Q3  |              |            | Q20 |              |            |
| Q4  |              |            | Q21 |              |            |
| Q5  |              |            | Q22 |              |            |
| Q6  |              |            | Q23 |              |            |
| Q7  |              |            | Q24 |              |            |
| Q8  |              |            | Q25 |              |            |
| Q9  |              |            | Q26 |              |            |
| Q10 |              |            | Q27 |              |            |
| Q11 |              |            | Q28 |              |            |
| Q12 |              |            | Q29 |              |            |
| Q13 |              |            | Q30 |              |            |
| Q14 |              |            | Q31 |              |            |
| Q15 |              |            | Q32 |              |            |
| Q16 |              |            | Q33 |              |            |
| Q17 |              |            | Q34 |              |            |

Please answer these additional questions:

- (a) Which was the hardest for you, among those you have solved? \_\_\_\_\_
- (b) Which seemed to be the hardest for you, among those you haven't solved? \_\_\_\_\_
- (c) Which was your favorite question, if you had to choose one? \_\_\_\_\_